DenseNet & CBAM

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DenseNet CBAM 실험 결과

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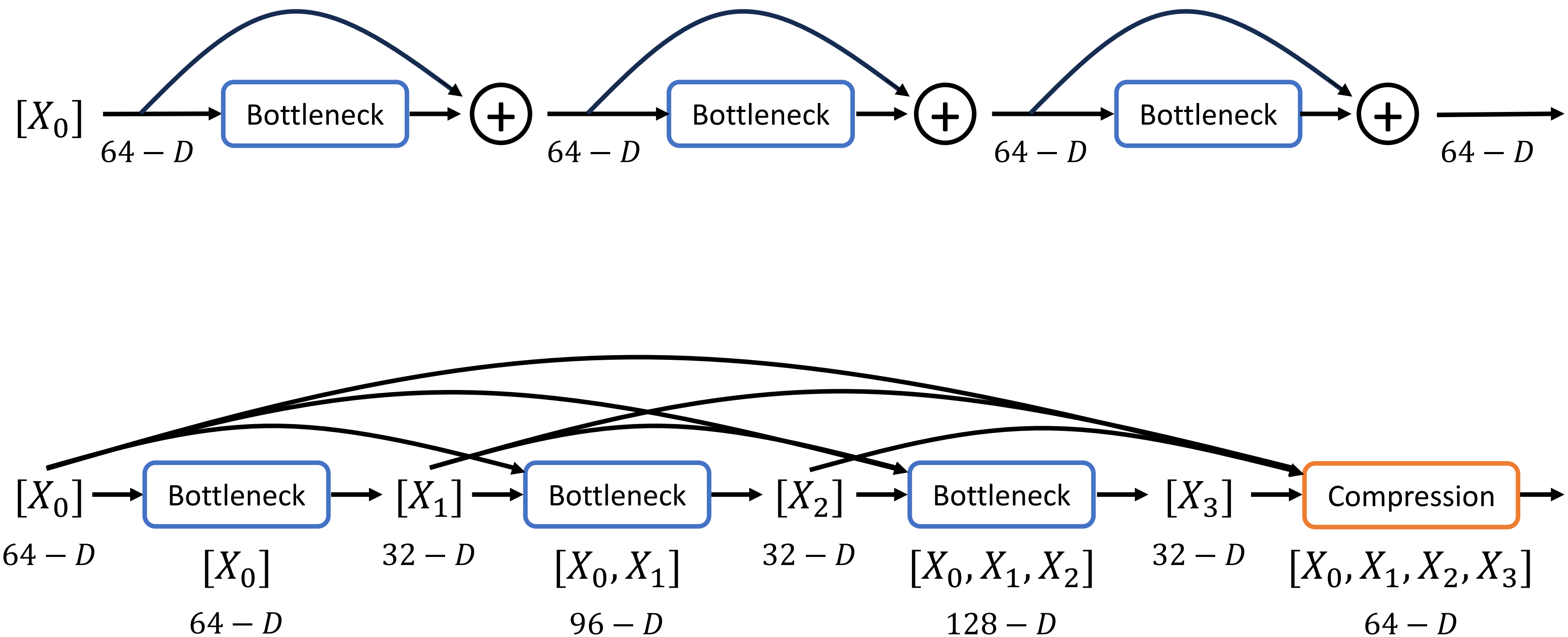
DenseNet[1]

- **Differences from ResNet**
  - ResNet의 Skip connection은 Summation, DenseNet은 Concatenation



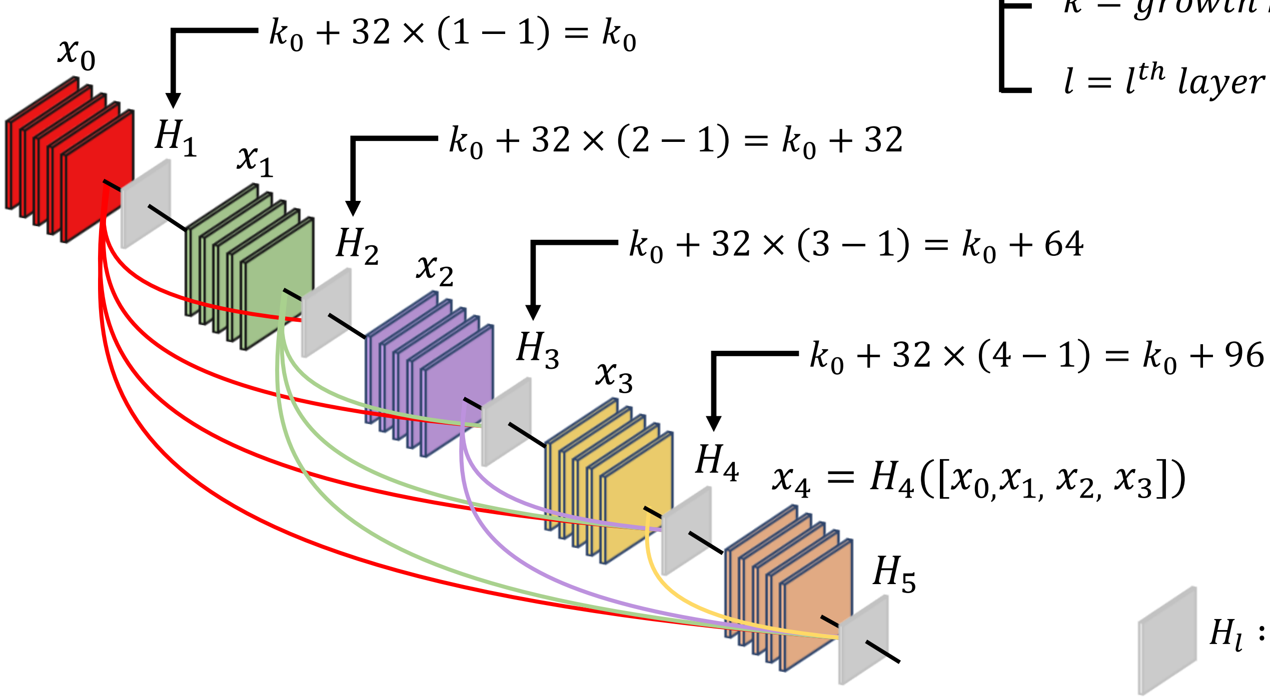
[1] G.Huang, Z.Liu et al. "Densely connected convolutional networks." In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4700-4708), 2017.

DenseNet

- **Main Idea**
  - Densely connected

$k_0 + k \times (l - 1) = l^{th}\ layer\ input\ feature\ map$

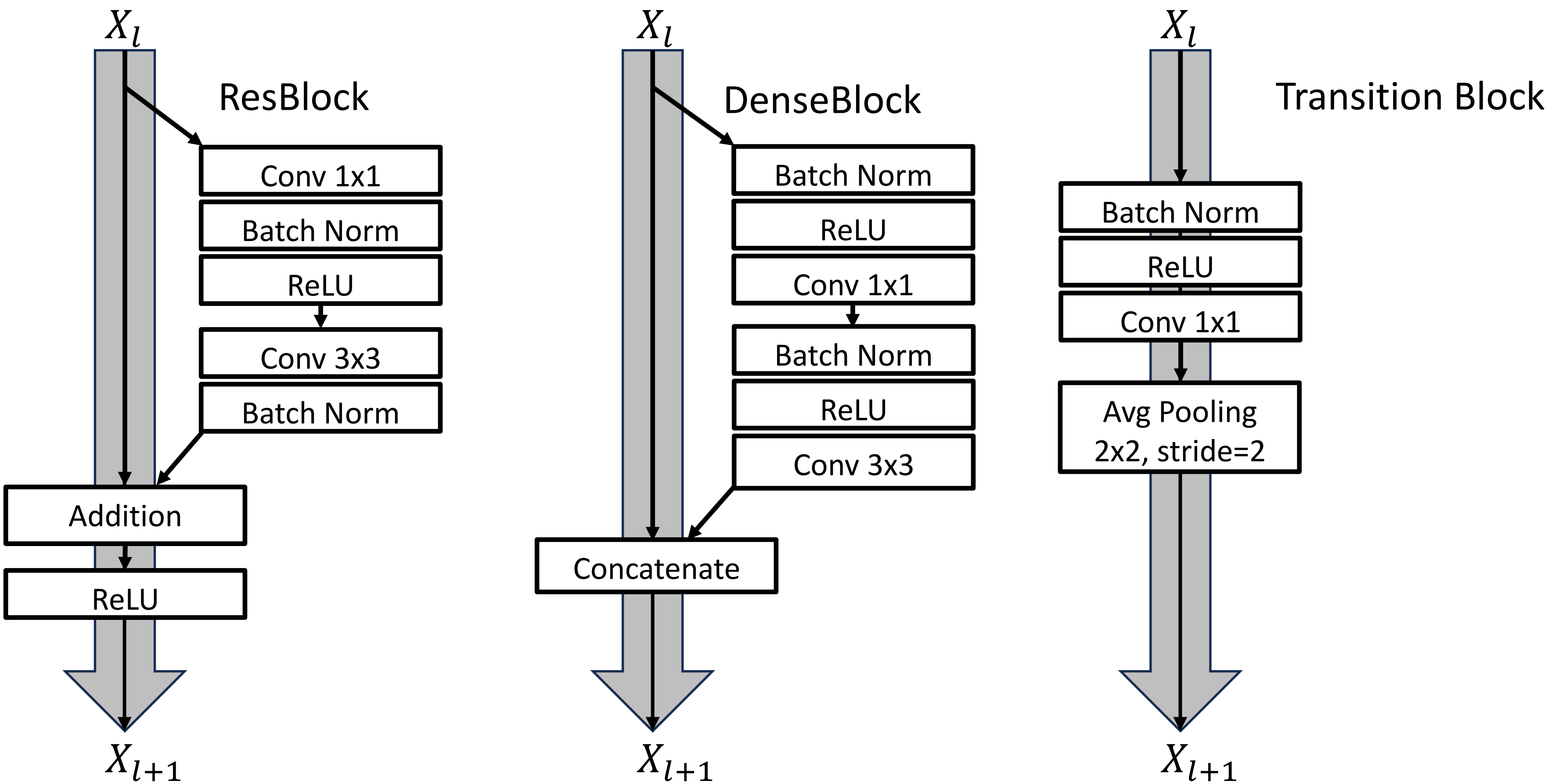
- $k_0 = number\ of\ channels\ in\ the\ input\ layer$
- $k = growth\ rate(32)$
- $l = l^{th}\ layer$



H_l : Non-Linear transformation

DenseNet

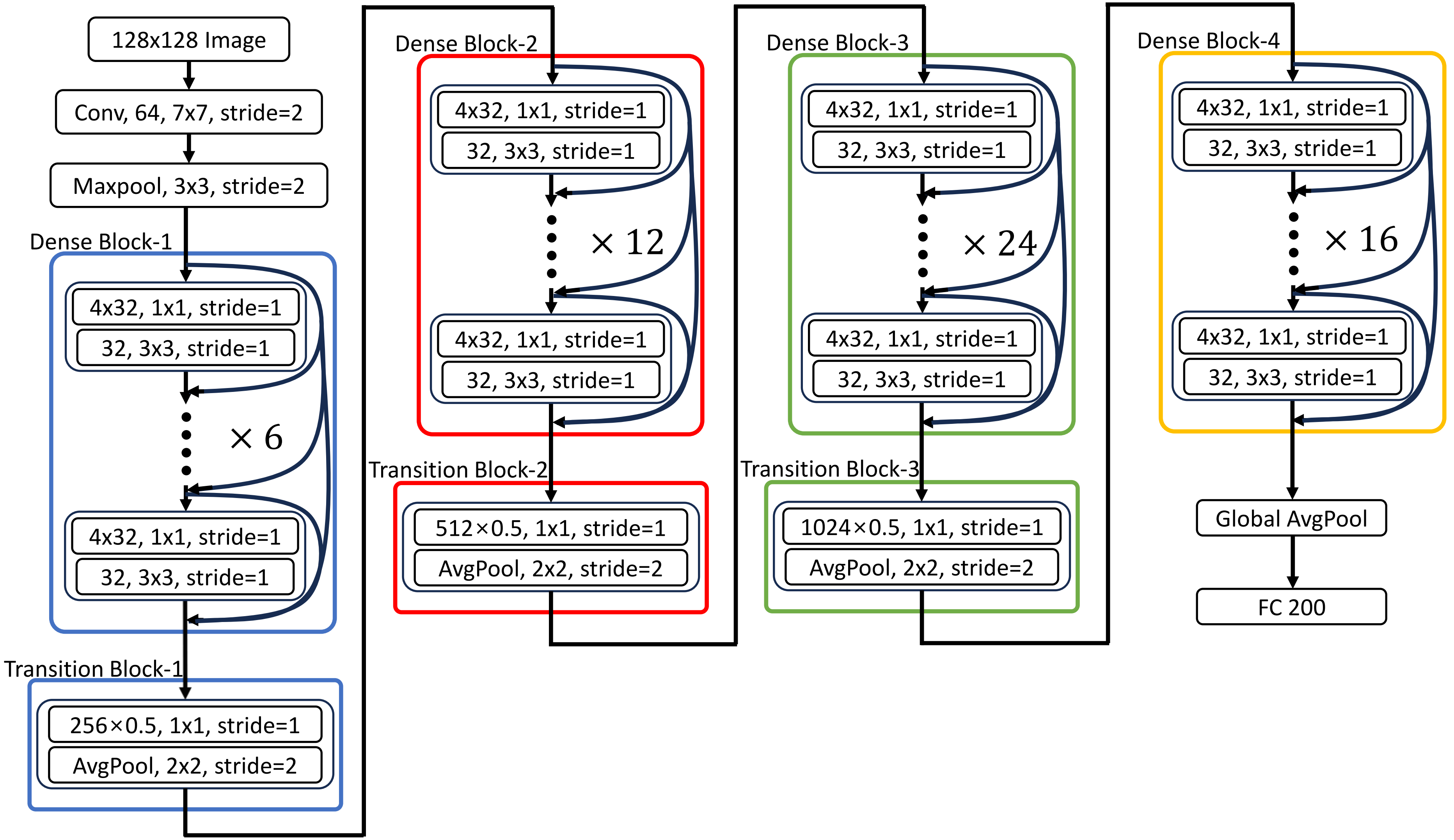
- **Main Idea**
  - Pre-activation[2]



[2] K.He, X.Zhang et al. “Identity mappings in deep residual networks.” In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, Proceedings, Part IV 14* (pp. 630-645). Springer International Publishing. 2016.

DenseNet

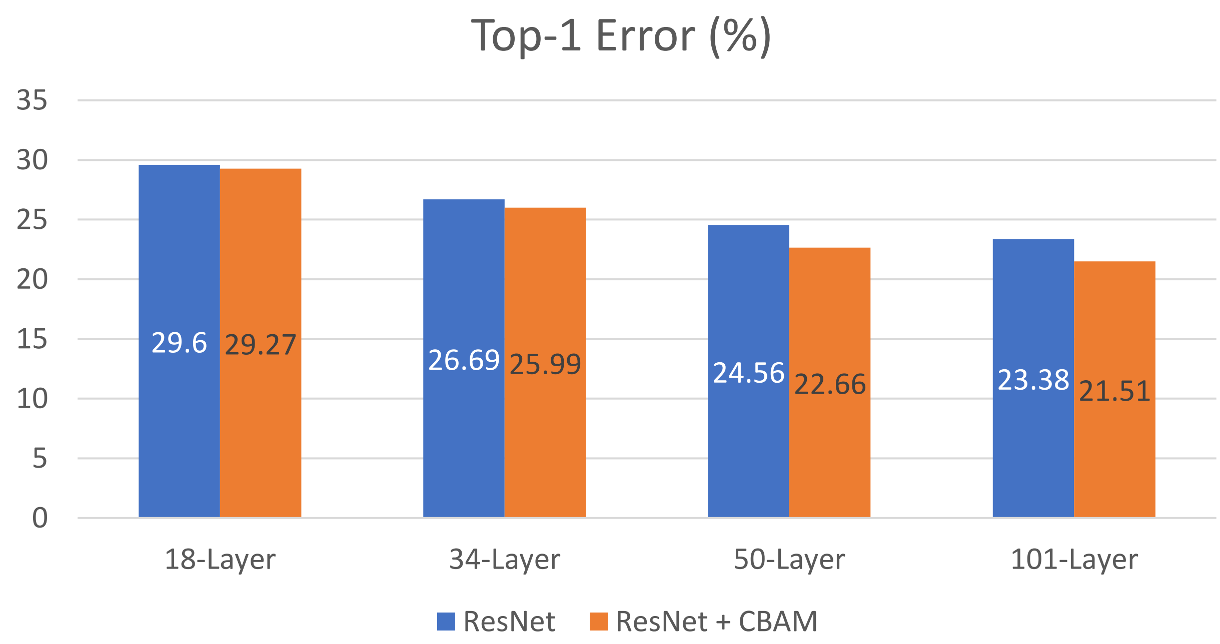
- **Architecture**



CBAM[3]

▪ Introduction

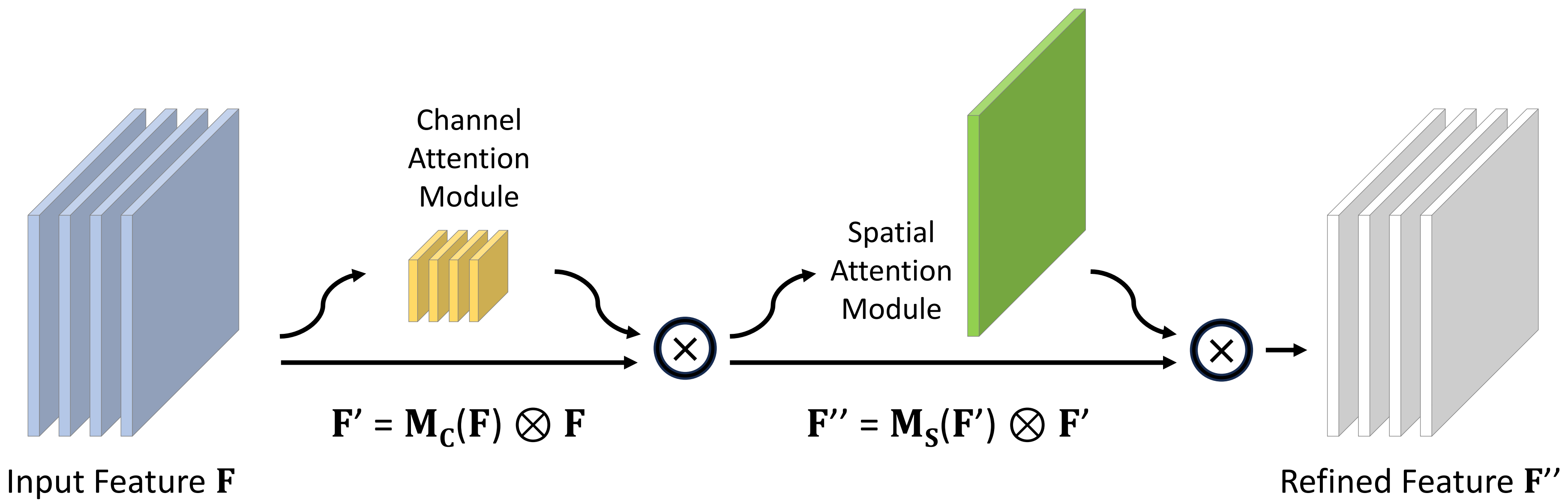
- CNN based model의 feature에 attention을 사용하여 더 의미있는 정보 추출
- Feature map의 중요한 정보는 강조, 중요하지 않은 정보는 억제
- 추가되는 parameter가 적어 계산량 부담이 줄어듬



[3] S.Woo, J.Park, J. Lee et al. “Cbam: Convolutional block attention module.” *In Proceedings of the European conference on computer vision (ECCV)* (pp. 3-19). 2018

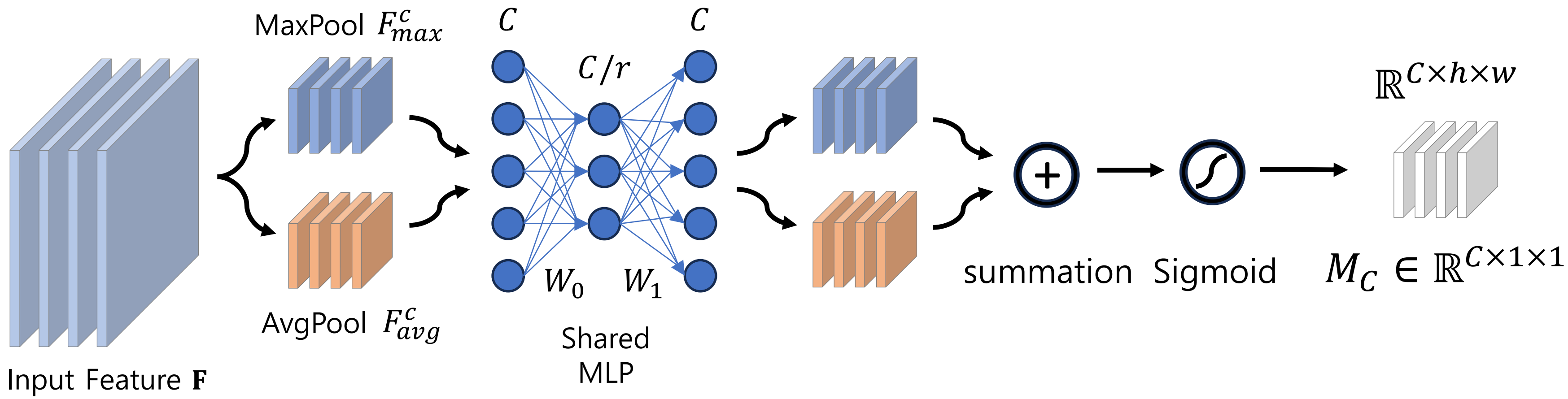
CBAM

- **Convolution Block Attention Module**
  - 두 개의 연속적인 Channel, Spatial attention Module 사용
  - 어떤 정보를 강조하거나 억제할지를 학습하면서 정보 흐름을 도움



CBAM

- **Channel Attention Module**
  - Input image에서 “무엇”이 의미 있는지에 집중
  - Channel attention을 효율적으로 계산하기 위해 spatial dimension을 압축



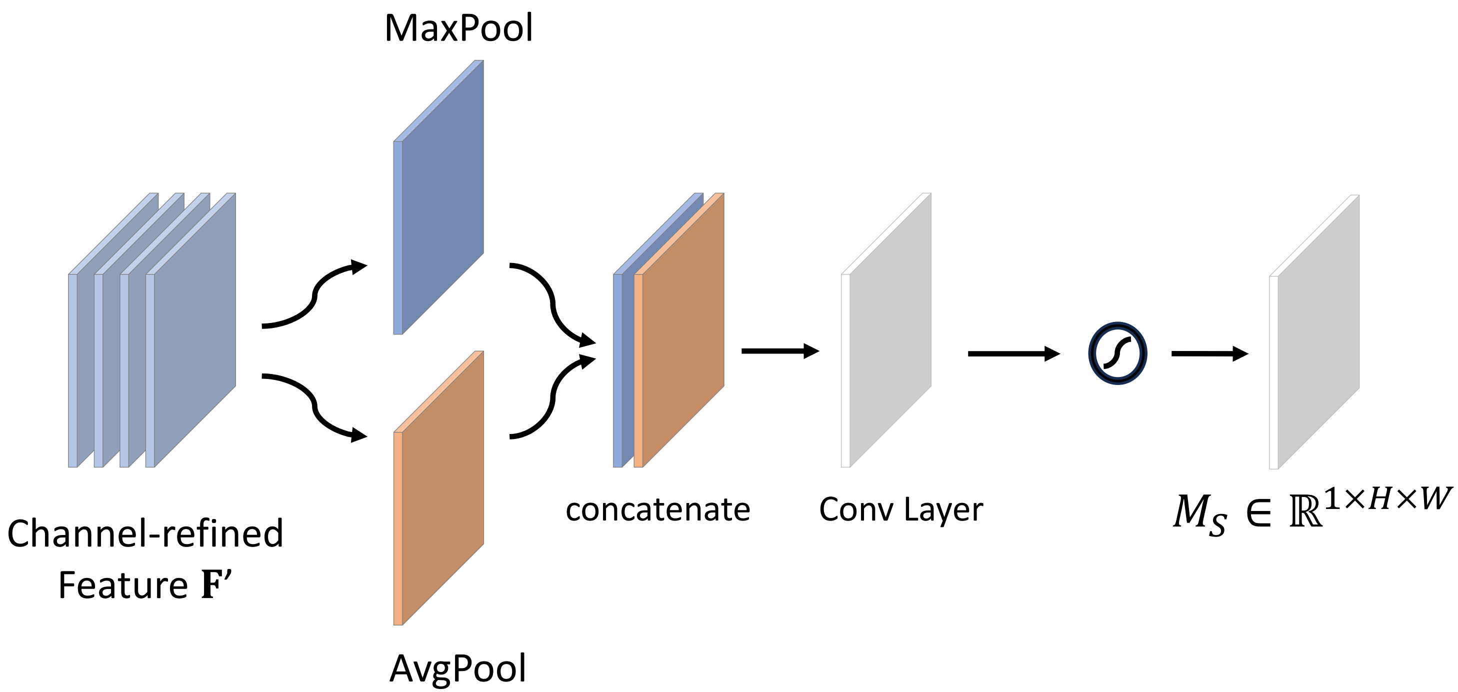
$$M_C = \sigma\left(MLP\big(AvgPool(F)\big) + MLP\big(MaxPool(F)\big)\right)$$

$$M_C = \sigma\left(W_1\left(W_0\big(F^c_{avg}\big)\right) + W_1\big(W_0(F^c_{max})\big)\right)$$

$W_1 \in \mathbb{R}^{C/r\times C}$ $\quad$ $W_0 \in \mathbb{R}^{C\times C/r}$ $\quad$ $r = reduction\ ratio$

CBAM

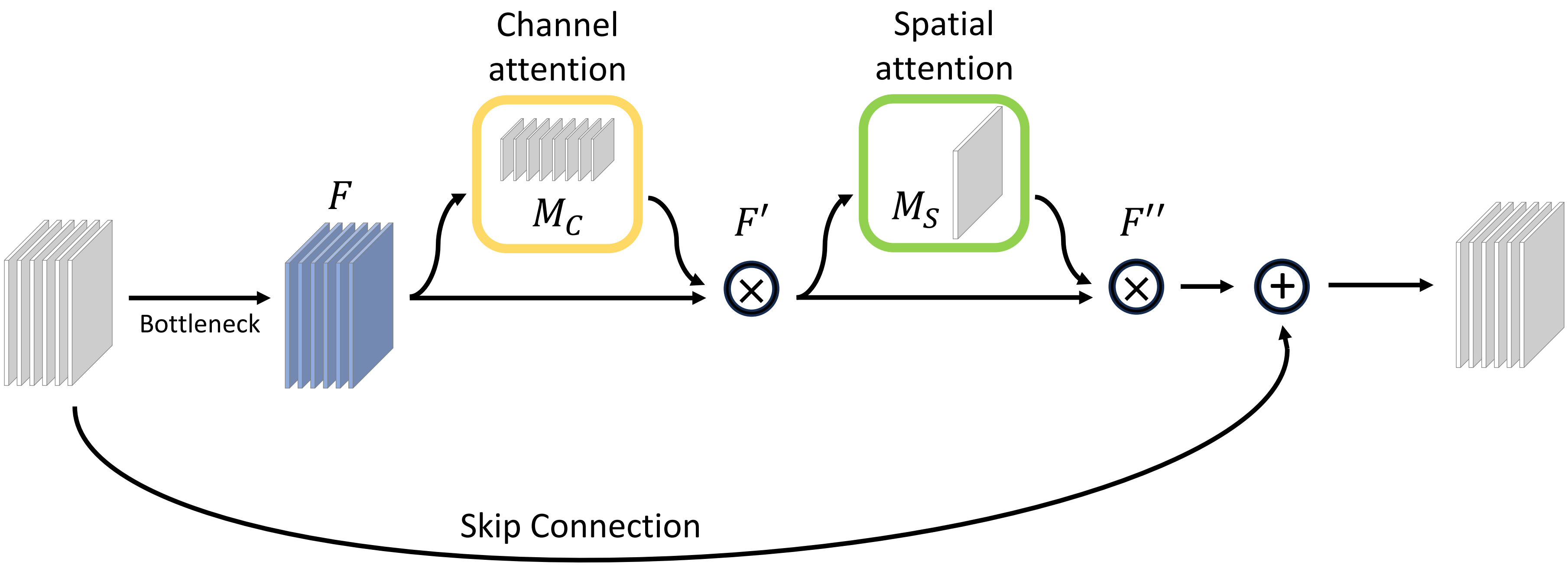
- **Spatial Attention Module**
  - Input image에서 “어디”가 의미 있는지에 집중
  - Channel 방향으로 Max, Average Pooling 진행



$$M_S(F) = \sigma\left(f^{7\times7}([AvgPool(F), MaxPool(F)])\right)$$

CBAM

- **ResBlock + CBAM**
  - Bottleneck의 Output feature를 Attention Module로 전달 후 Skip Connection



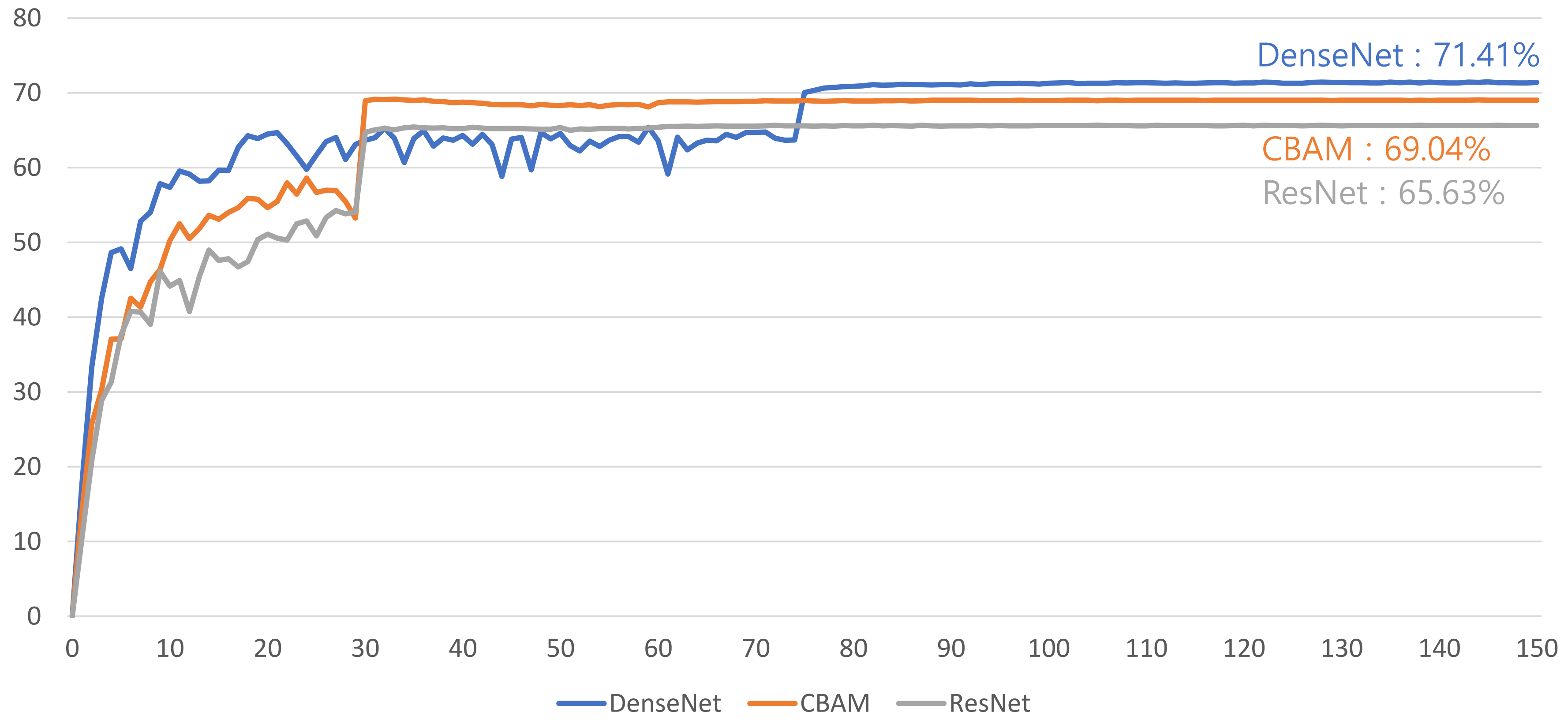
Experiment

- **DenseNet & ResNet+CBAM settings**
  - DataSet
    - Image : TinyImageNet
    - Size : 128 × 128
    - Train : 207,005
    - Test : 51,752
    - Class : 200
  - Augmentation
    - Random Crop
    - Random Horizontal Flip
  - DenseNet HyperParameter
    - Batch size : 64
    - Epoch : 150
    - Scheduling : 50%, 75%, 0.1
    - Learning rate : 0.01
    - Loss Function : Cross Entropy
    - Optimizer : SGD
    - Momentum : 0.9
    - Weight decay : 0.0001
  - CBAM HyperParameter
    - Batch size : 256
    - Epoch : 150
    - Scheduling : Epoch 30마다 0.1
    - Learning rate : 0.01
    - Loss Function : Cross Entropy
    - Optimizer : SGD
    - Momentum : 0.9
    - Weight decay : 0.0005

Result

▪ DenseNet & ResNet+CBAM

- DenseNet : 71.41%
- ResNet : 65.63%
- ResNet+CBAM : 69.04%



The end

Thanks

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